Work?

Part 2: Neural Networks

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## Review of Previous Presentation

*Previous Presentation on How Does AI Work? Part 1: Pattern Recognition here.*

**Input $\rightarrow$ Machine Learning Model $\rightarrow$ Output.**

Machine Learning models utilize pattern recognition to predict and classify data inputs.

A simple type of pattern recognition is Nearest Neighbor, where input is categorized based on its proximity to existing data points.

Neural Networks arefundamentally

**still a pattern recognition**

system.

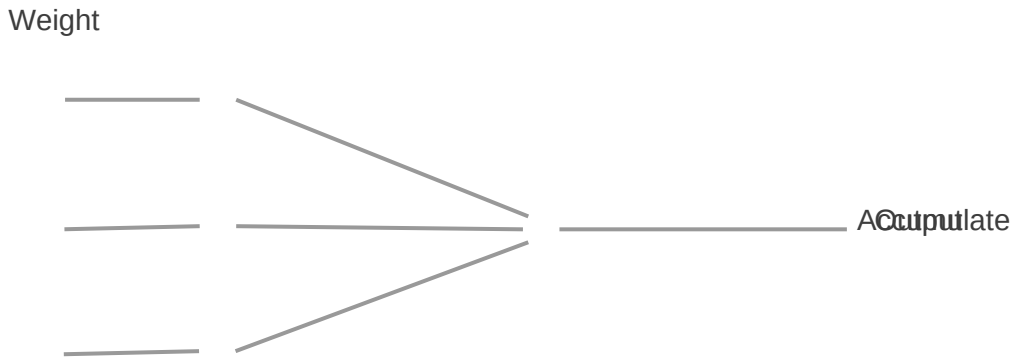
Neural Networks (cont.)

A neural network is a type of ML model that consists of a set of connected nodes where each node assigns weights to its inputs.

Node: a basic building block of neural networks that assigns weights to inputs.

Weight: a set of numbers that make up the model.

E.g. list of words/sentences in the contract

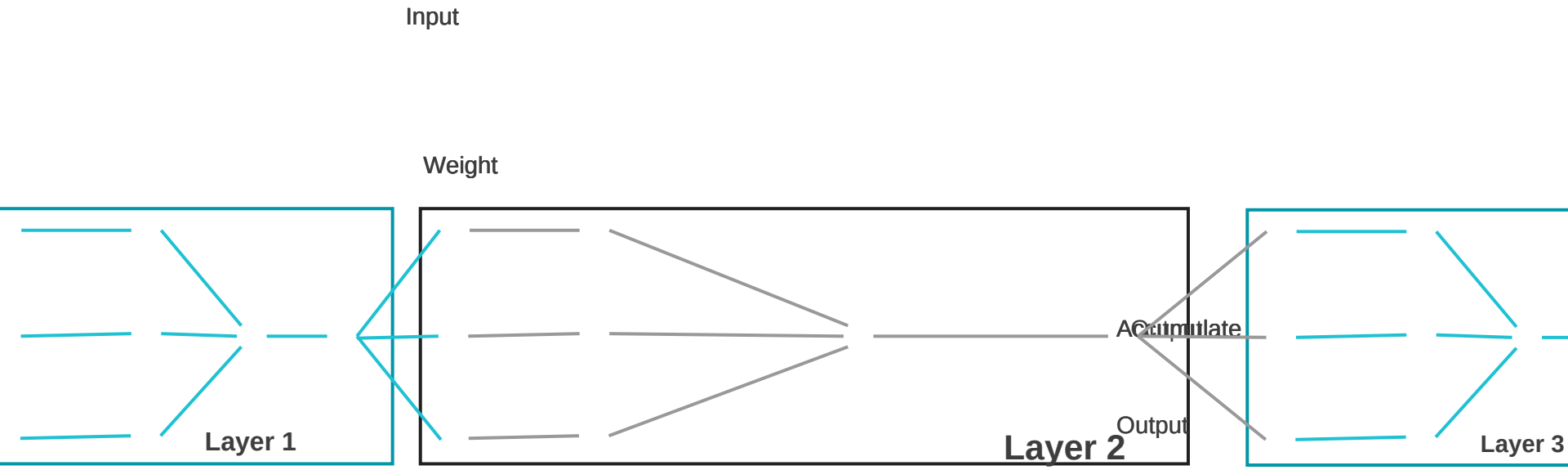


# From Neural Networks to Deep Learning

"Deep" learning refers to a type of ML model that uses layers of neural networks.

This allows it to make more complex decisions, like speech-to-text, sentiment detection, and text summarization.

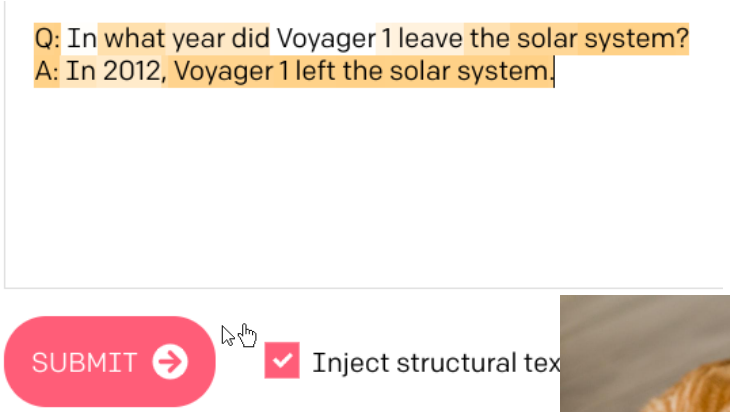
Nowadays, almost all models have multiple layers, and some have hundreds of layers.

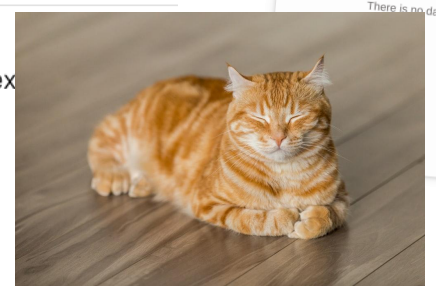


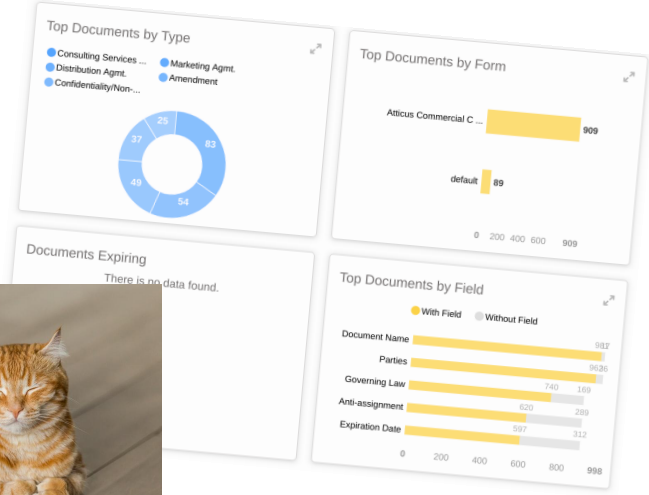
# Output

The AI will then produce an output in a format that humans can understand.
A simple output would be the probability of a clause being a Change of Control clause.

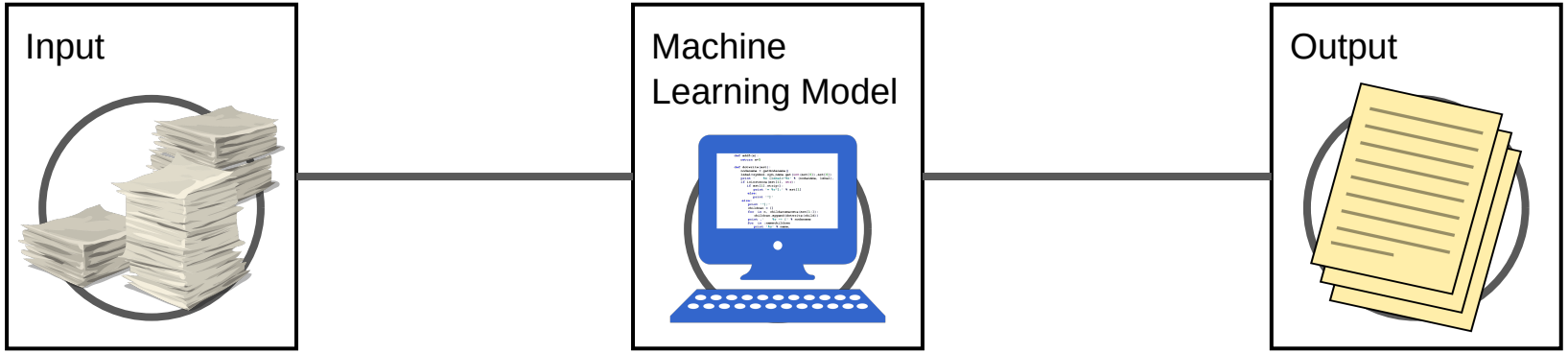
More advanced outputs, such as long text, images, and graphs, etc., require additional processing.





Review of Steps




Pre-processing, usually done using a simpler program.

Based on Pattern Recognition

Neural Networks: made of systems of nodes, communicating with each other

Deep Learning: Neural Networks with multiple layers of processing

AI produces an output humans can use and understand.